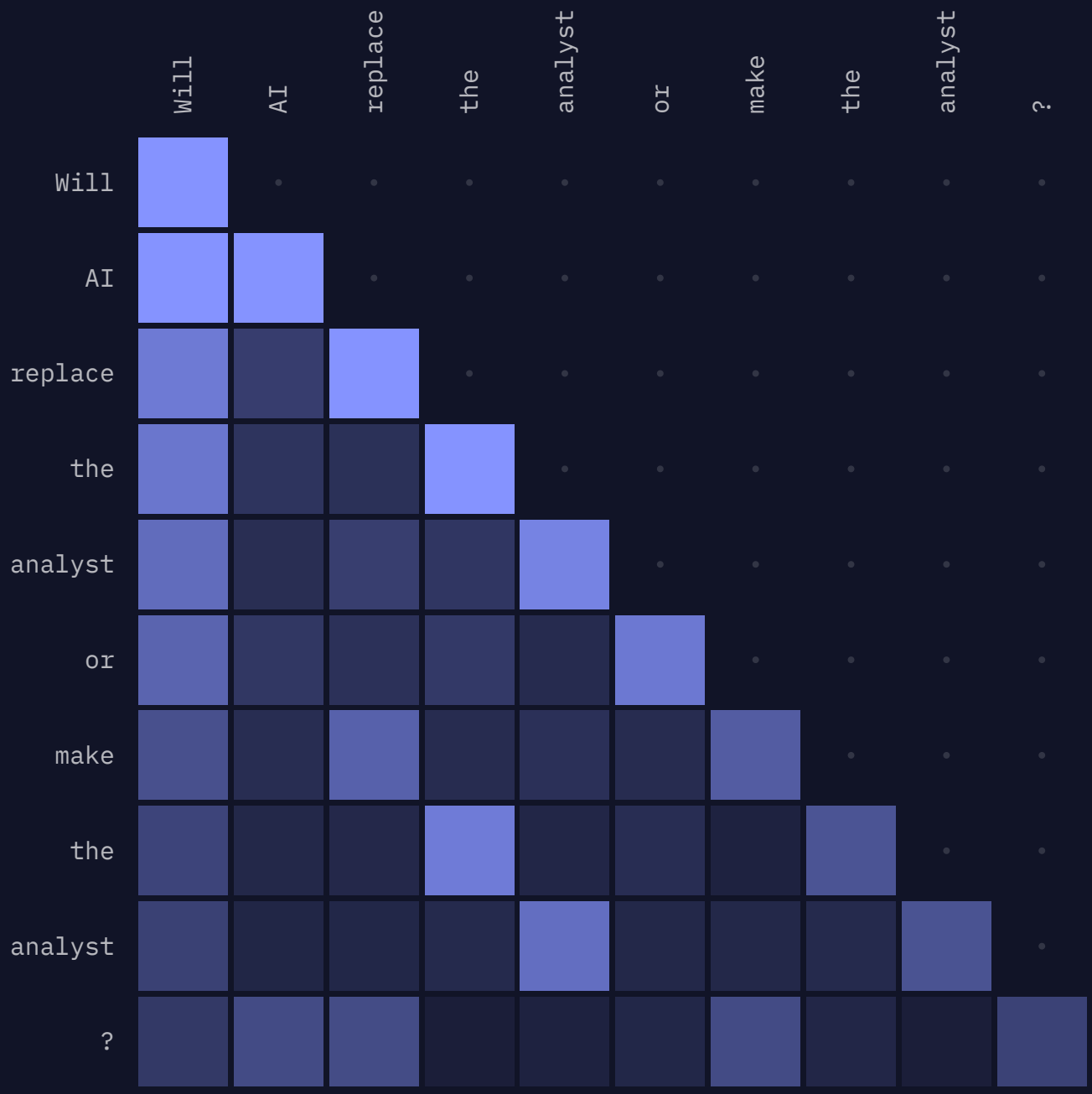
INFERENCE.



The AI compendium for the business mind

BEFORE YOU BEGIN

Three ways to read this book

This compendium is dense on purpose. It is also built so that you never have to read all of it at once. Pick the speed that matches how much time is left before your interview.

THE NIGHT BEFORE

Skim

Read only the highlighted lines and the yellow **night-before page** at the end of each chapter. About ten minutes a chapter.

THE WEEK BEFORE

Study

Read the full text. Ideas are built from first principles, so read chapters in order the first time. Notes and definitions sit in the outer margin.

EVERY DAY

Practise

Answer **Check yourself** without looking, then run the **Try it yourself** prompts in a real AI tool. Recall beats rereading.

WHAT THE BLOCKS MEAN

∴ FIRST PRINCIPLES	An idea derived step by step from things you already know. Graph paper means: work it through, don't memorise it.
CASE	A business decision worked end to end. Marked *Illustrative* when it is constructed for teaching.
• INTERVIEW ROOM	A question you could be asked, what it tests, and the structure of a strong answer.
CONCEPT CARD	One term, in plain words, with an analogy and the most common misconception.
■ FIELD LENS	The same idea through one placement track. The colour matches the thumb tab on the page edge.
WATCH · READ	Curated outside material from named, primary sources. Scan the QR code on the printed page.

CONTENTS

What's inside

I How the machine thinks

First principles of AI for people who will never write a line of code.

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II AI at work, by function

The same ideas, seen through the six placement lenses.

04	The consultant's new toolkit	planned ·
05	Marketing when content is free	planned ·
06	Building AI products	planned ·

III The interview room

Cases, guesstimates and GD topics where AI is the subject.

07	AI cases, worked end to end	planned ·

READING PATHS BY PLACEMENT TRACK

CONSULTING

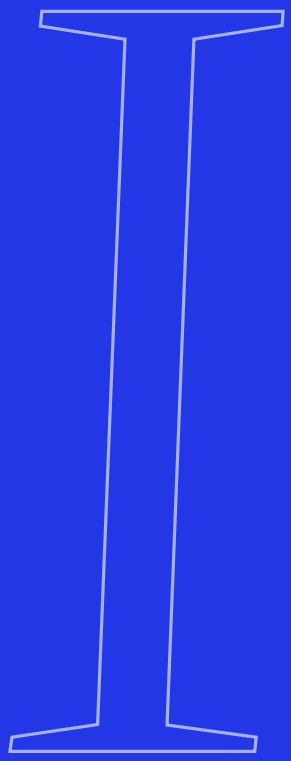
1. 01 · What is a model, really?
2. 02 · Next-token prediction, and why it looks like thinking
3. 03 · Where the machine breaks
4. 04 · The consultant's new toolkit

MARKETING

1. 01 · What is a model, really?
2. 02 · Next-token prediction, and why it looks like thinking
3. 03 · Where the machine breaks
4. 05 · Marketing when content is free

PRODUCT

1. 01 · What is a model, really?
2. 02 · Next-token prediction, and why it looks like thinking
3. 03 · Where the machine breaks
4. 06 · Building AI products



PART I

How the machine thinks

First principles of AI for people who will never write a line of code.

01 What is a model, really?

02 Next-token prediction, and why it looks like thinking

03 Where the machine breaks

01

What is a model, really?

Every AI conversation in a placement interview rests on one idea. Get it in your bones and the rest of this book becomes reasoning, not memorisation.

FULL READ
9 min

SKIM LAYER
3 min

LEVEL
Zero to one

LENSES
FOUNDATIONS · CONSULTING · MARKETING · OPERATIONS

THE PUZZLE

Your food-delivery app's ETA is a guess. So is every word ChatGPT writes. Explain, in one sentence, what the two guesses have in common.

BY THE END OF THIS CHAPTER YOU CAN

1. **Define** a model as a learned rule from inputs to outputs, without using the word "algorithm".
2. **Distinguish** training from inference, and explain why one costs crores and the other costs paise.
3. **Explain** a large language model as the same idea as a delivery-time model, at a different scale.
4. **Spot** the three questions to ask before trusting any model a client wants to buy.

FND

Before the word "AI" meant anything to you, you were already surrounded by models. The cricket commentator who says a target of 180 is "par on this pitch" is running one. So is the shopkeeper who orders more milk before Diwali. A model is a rule that turns what you know into a guess about what you don't. Everything in this book, from chatbots to credit scoring, is a version of that sentence.

This chapter builds the idea from nothing. We will start with a rule you could write on a napkin, make it learn from data, and then scale it up until it is a large language model. At no point will you need mathematics beyond multiplication.

§ 01.1

A model is a compressed rule

Take a delivery app predicting how long your order will take. The app knows a few things at the moment you order: the distance to the restaurant, the time of day, whether it is raining. It does not know the one thing you care about, which is when the food arrives. A model is the bridge between the two.[1]

The simplest bridge is a line. Suppose that, looking at last month's orders, every extra kilometre added roughly four minutes. You could write the rule as:

delivery time = 12 *minutes* + 4 × *distance in km*

That is a model. The numbers 12 and 4 are its parameters: the knobs that make this rule fit this city. Change the city and the knobs change; the shape of the rule stays.

1 A **forecast**, a **score** and a **recommendation** are all the same object: a model's output. Only the business use differs.

Parameter
One adjustable number inside a model (a "weight"). Training sets its value. Large language models have billions.

∴ FROM FIRST PRINCIPLES

Build it from zero

- **P1** We want a number we cannot see yet (delivery time) from numbers we can see now (distance).
- **P2** Pick a rule with adjustable knobs: *time* = *a* + *b* × *distance*.
- **P3** Guess the knobs, check the guess against real past orders, and measure how wrong it was.
- **P4** Nudge the knobs in whichever direction reduces the error. Repeat thousands of times.
- **∴** What you are left with is a rule that summarises thousands of orders in two numbers. A model is a compression of experience into parameters.

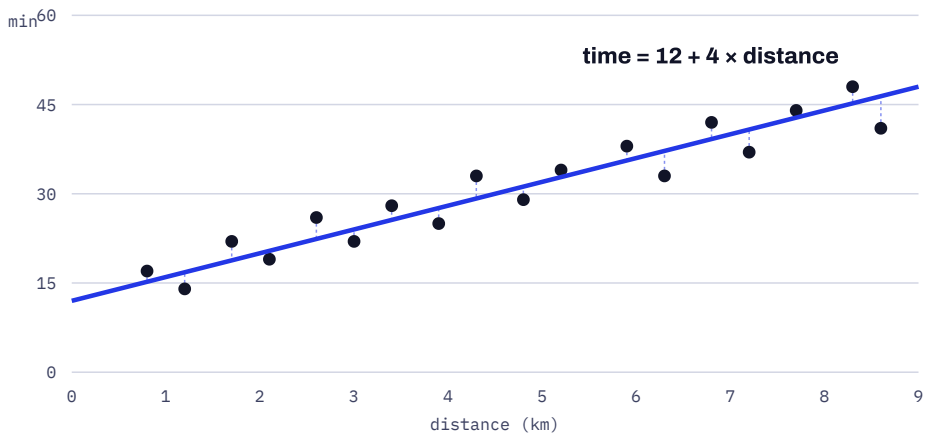


FIG 01.1 Each dot is a past order. Training is the search for the line that misses the dots by the least. The line is the model; its slope and starting point are its two parameters.

Illustrative data

§ 01.2

Training and inference are different businesses

The process of nudging the knobs is called training. Using the finished rule on a new order is called inference. The distinction sounds academic. It is the single most important cost fact in AI economics.

Training is done once, or occasionally, and it is expensive because the model has to look at every example many times. Inference is done every time anyone uses the model, and each use is cheap. For our two-parameter line, both are trivial. For a frontier language model, training runs into hundreds of crores of compute; a single answer costs a fraction of a rupee.[2]

	TRAINING	INFERENCE
What happens	Knobs are set from examples	Set knobs are used on a new input
How often	Once, then periodic refreshes	Every query, millions of times a day
Cost shape	Large, upfront, fixed	Small, per use, variable
Business analogy	Building the factory	Running the production line

Training

Adjusting a model's parameters, example after example, to shrink its error. Expensive, done once.

Inference

Using a trained model to produce an output. Cheap per use, done millions of times. The book is named after it.

2 The Stanford AI Index tracks both numbers year by year [AI, 2025]. Quote it, not the press coverage of it.

FND

CONCEPT 01.1

Model

PLAIN WORDS. A rule, learned from examples, that turns an input into a guess about an output.

ANALOGY. A seasoned credit officer who has seen ten thousand loan files. They cannot recite every file, but the files have shaped their judgement. The judgement is the model.

MISCONCEPTION. "The model looks up the answer." It does not. After training, the examples are gone; only the parameters remain. This is why a model can be confidently wrong about something it has never seen.

§ 01.3

From two knobs to billions

A delivery-time model with two parameters can only learn a straight line. Give it more inputs (rain, traffic, restaurant prep time) and more knobs, and it can learn a more detailed rule. Give it millions of knobs arranged in layers, and it can learn rules no human could write down, such as "what does a cat look like" from raw pixels.

Large language model (LLM)
A model trained to predict the next token of text, at a scale that makes it useful for writing, reasoning and conversation.

Token
The unit of text a language model reads and writes, roughly three-quarters of an English word.

A large language model is this same machine, pointed at text. Its input is all the text so far. Its output is a guess about the next token, the next small piece of a word. It has billions of parameters, set by training on a vast amount of written text. When you ask it a question, it guesses one token, appends it, and guesses again, token after token, until it has written an answer.

> "*All models are wrong, but some are useful.*
>
> GEORGE E. P. BOX, STATISTICIAN

That loop is the whole trick. There is no separate "understanding" module and no database of facts; there is one very large rule for continuing text. The surprise of the last few years is how much useful behaviour falls out of continuing text well. Writing a good continuation of a question about a balance sheet requires something that looks a lot like reasoning about the balance sheet.

▸ WATCH

60 min

Tier A · primary source

[1hr Talk] Intro to Large Language Models

Andrej Karpathy · YouTube · 2023

A founding member of OpenAI explains, for a general audience, what an LLM is physically (two files) and how it is trained. The single best hour you can spend on this chapter.

- 0:20 An LLM is two files: parameters + code
- 4:17 Training as lossy compression of the internet
- 14:14 From document generator to assistant

youtube.com

▸ WATCH

19 min

Tier A · primary source

But what is a neural network?

Grant Sanderson · 3Blue1Brown · 2017

The clearest visual explanation of what "parameters" and "layers" are. Watch it before the next chapter and the word "weights" will never be abstract again.

youtube.com

MYTH / REALITY	
~~MYTH.~~ An LLM searches the internet or a database for the right answer, then rewrites it.	REALITY. A plain LLM generates every answer token by token from its parameters. Some products add a search step on top, but the model itself retrieves nothing. That is why it can invent a citation that looks perfect.

§ 01.4

Three questions before you trust any model

Because every model is a compression of past examples, three questions expose most of its risk. Use them in any case interview where a client wants to "use AI".

1. **What examples was it trained on?** A model is only as good as the experience it compressed. A delivery model trained in dry Bengaluru will be wrong in a Mumbai monsoon.
2. **What happens when it is wrong?** A bad song recommendation costs nothing. A bad credit decision costs money and invites the regulator. Error

FND

cost decides how much human review you need.

3. **Has the world changed since training?** Models learn the past. If prices, customers or rules have shifted, the model is describing a world that no longer exists.[3]

3 Statisticians call this *distribution shift*. You do not need the term in an interview; you need the question.

40%

Higher quality output from BCG consultants using GPT-4, on tasks inside the AI's capability. On a task designed to sit outside it, AI users were 19 percentage points *less* likely to get the right answer.

Source: [Dell'Acqua et al., 2023]

The BCG experiment is worth knowing by heart, because it shows the third question in action. AI made consultants faster and better on tasks it was good at, and worse on a task that looked similar but needed judgement it lacked. The researchers called the boundary between the two a *jagged frontier*: you cannot see where it is from the outside.

READ

12 min

Tier B · established analysis

oneusefulthing.org

Centaurs and Cyborgs on the Jagged Frontier

Ethan Mollick · One Useful Thing · 2023

A co-author of the BCG study explains its findings in plain language. Read this if you will not read the paper.

CASE

Illustrative · A quick-commerce dark store

THE SITUATION

A 10-minute delivery company stocks each dark store from a demand forecast, a model that predicts tomorrow's sales of each item from past sales, weather and day of the week. Waste on fresh items is running at 9% and the category head wants "a better AI model".

THE DECISION

Before buying a bigger model, ask the three questions. What was it trained on? Two years of data, including lockdown months that no longer describe demand. What happens when it is wrong? Over-forecasting milk is waste; under-forecasting is a lost order and a churned customer, and those costs are not equal. Has the world changed? A competitor opened next door in March.

YOUR MOVE

The fix may be cheaper than a new model: drop the stale months from training, and change the target from "most likely demand" to "the stock level that balances the cost of waste against the cost of a stockout". That is a business decision, not a technical one, and it is exactly the kind of decision recruiters want you to be able to make.

CONSULTING LENS

Clients rarely ask “should we build a model?” They ask “where can AI cut cost?” Translate it: which decisions are made often, on data we already have, where a wrong guess is cheap? That set is where models pay back first.

MARKETING LENS

Every targeting, churn and lifetime-value score you will use is a model. Ask what it was trained on before you trust a segment it hands you.

OPERATIONS LENS

Demand forecasting is the oldest business model of all. The new tools change the inputs (text, images, weather feeds), not the logic of trading off waste against stockouts.

• INTERVIEW ROOM

Guesstimate meets AI

Q. A mid-size NBFC wants to “use AI for loan approvals”. You have two minutes. Where do you start?

> **“Before the model, the decision: what does a wrong approval cost versus a wrong rejection, and how many decisions a day are we talking about?”**

What they are testing. Whether you treat AI as a business decision with error costs, or as magic.

A strong structure. Decision and volume → cost of each kind of error → data the NBFC already has → regulatory constraints on explaining a rejection → start with human-in-the-loop on the riskiest band.

▸ TRY IT YOURSELF

Watch a model guess

Open any chat assistant and paste this three times, in three new chats:

```
Complete this sentence with exactly five more words, nothing else:
"The best thing about Indian monsoon is"
```

You will probably get different endings. The model is sampling from its guesses about the next token, not retrieving a fixed answer. Now ask it why it chose its ending, and notice that the explanation is also just more generated text.

FND

READ

90 min

Tier A · primary source

What Is ChatGPT Doing … and Why Does It Work?

Stephen Wolfram · Stephen Wolfram Writings · 2023

Long, patient and illustrated. Read the first three sections ("It's just adding one word at a time" through "Where do the probabilities come from?") and stop there for now.

writings.stephenwolfram.com

RETRIEVAL PRACTICE

Check yourself

- ☐ Q1 In one sentence, without the word “algorithm”, what is a model? `recall`
- ☐ Q2 A model has been trained. Is answering one customer query training or inference? Which one dominates the cost of a free chatbot with 10 crore users? `apply`
- ☐ Q3 Why can an LLM produce a convincing but invented citation? Use the word “parameters” in your answer. `explain`
- ☐ Q4 A bank’s fraud model was trained before UPI took off. Which of the three trust questions is most urgent, and why? `transfer`
- ☐ Q5 Give one business decision where you would *not* let a model act without a human, and say what makes the error cost high. `judge`

ANSWERS — COVER THEM, TRY FIRST

A1 A rule, learned from past examples, that turns information you have into a guess about something you don’t.

A2 Inference. For a heavily used product, inference costs, paid on every query, can exceed the one-time training cost.

A3 Its parameters store patterns of text, not a list of sources. It generates a string that has the shape of a citation without retrieving one.

A4 “Has the world changed since training?” Payment behaviour shifted after UPI, so the model is describing old fraud patterns.

A5 Any decision with a high or irreversible error cost, such as rejecting a loan or a medical triage call, especially where you must explain the decision to a regulator.

SKIM LAYER

The night-before page

01 A model is a learned rule: inputs you have → a guess about an output you don't.

02 Parameters are the knobs; training sets them from examples; the examples are then discarded.

03 Training is the factory (big, upfront); inference is the production line (small, per use).

04 An LLM is a next-token guesser with billions of parameters. Useful behaviour comes from guessing text well.

05 Before trusting any model ask: trained on what, costly when wrong how, and has the world changed?

BACK MATTER

Glossary

Every term defined in the margins, in one place. Learn these words well enough to explain them to a non-technical interviewer in one sentence each.

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Using a trained model to produce an output. Cheap per use, done millions of times. The book is named after it.

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Sources

Every outside source in this edition, with its tier. **A** is a primary source or originator, **B** is established editorial or analysis, **C** is admitted with a stated reason. No SEO listicles, no unsourced posts. Rules in the repository's SOURCES.md.

A **[1hr Talk] Intro to Large Language Models**
Andrej Karpathy · YouTube · 2023
https://www.youtube.com/watch?v=zjkBMFhNj_g

A **But what is a neural network?**
Grant Sanderson · 3Blue1Brown · 2017
https://www.youtube.com/watch?v=aircAruvnKk

B **Centaurs and Cyborgs on the Jagged Frontier**
Ethan Mollick · One Useful Thing · 2023
https://www.oneusefulthing.org/p/centaurs-and-cyborgs-on-the-jagged

A **What Is ChatGPT Doing … and Why Does It Work?**
Stephen Wolfram · Stephen Wolfram Writings · 2023
https://writings.stephenwolfram.com/2023/02/what-is-chatgpt-doing-and-why-does-it-work/

A **AI Index Report**
Stanford Institute for Human-Centered AI · Stanford HAI · 2025
https://hai.stanford.edu/ai-index

A **Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality**
Dell'Acqua, McFowland, Mollick, Lifshitz-Assaf, Kellogg, Rajendran, Krayer, Candelon, Lakhani · Harvard Business School Working Paper 24-013 · 2023
https://www.hbs.edu/faculty/Pages/item.aspx?num=64700

COLOPHON

Set in Archivo, Literata and IBM Plex Mono. Written in Markdown, laid out with Paged.js, rendered by Chromium. Placement Edition 2027 · Draft 0.1. Free to share with attribution. Not for sale.

Every B-school recruiter now asks about AI. Most compendiums still don't teach it. This one starts from first principles and ends in the interview room.